Topic: Initial value problems

Question: What can we interpret from an initial condition of $y(5) = 0$?

Answer choices:

A $C = 0$

B $x = 0$ and $y = 5$

C $C = 5$

D $x = 5$ and $y = 0$

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Solution: D

An initial condition for a first order differential equation will take the form $y(x_0) = y_0$. To use it, we substitute $x = x_0$ and $y = y_0$ into the general solution to find the associated value of C. In our case, $x = 5$ and $y = 0$.

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Topic: Initial value problems

Question: If $y(0) = -1$, solve the initial value problem.

$$y' + y = -x^2$$

Answer choices:

A $y = -x^2 + 2x - 2 + e^{-x}$

B $y = -x^2 + 2x - 2$

C $y = -x^2 + 2x - 2 - e^{-x}$

D $y = -x^2 - 2x + 2 - 3e^{-x}$

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Solution: A

The linear differential equation is already in standard form, so we can identify $P(x)$ and $Q(x)$.

$$P(x) = 1$$

$$Q(x) = -x^2$$

We use $P(x)$ to find the integrating factor.

$$\mu(x) = e^{\int P(x)\ dx}$$

$$\mu(x) = e^{\int 1\ dx}$$

$$\mu(x) = e^x$$

Multiply through the differential equation by the integrating factor.

$$e^x\left(\frac{dy}{dx} + y = -x^2\right)$$

$$e^x\frac{dy}{dx} + e^x y = -x^2 e^x$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(e^x y) = -x^2 e^x$$

then integrate, using integration by parts with $u = x^2$, $du = 2x\ dx$, $dv = e^x\ dx$, and $v = e^x$ to integrate the right side.

$$\int \frac{d}{dx}(e^x y) = -\int x^2 e^x\ dx$$

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$$e^x y = -\left(x^2 e^x - \int 2xe^x \, dx\right)$$

$$e^x y = -x^2 e^x + \int 2xe^x \, dx$$

Apply integration by parts a second time, this time with $u = 2x$, $du = 2 \, dx$, $dv = e^x \, dx$, and $v = e^x$.

$$e^x y = -x^2 e^x + \left(2xe^x - \int 2e^x \, dx\right)$$

$$e^x y = -x^2 e^x + 2xe^x - 2e^x + C$$

$$y = -x^2 + 2x - 2 + \frac{C}{e^x}$$

Once we have this general solution, we recognize from the initial condition $y(0) = -1$ that $x = 0$ and $y = -1$, so we'll plug these values into the general solution,

$$-1 = -0^2 + 2(0) - 2 + \frac{C}{e^0}$$

and then simplify to solve for C.

$$-1 = -2 + C$$

$$C = 1$$

So the solution is

$$y = -x^2 + 2x - 2 + \frac{1}{e^x}$$

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Topic: Initial value problems

Question: A function $y(x)$ is a solution of $y' + ky = 0$. Suppose that $y(0) = 1$ and $y(1) = e$. Find the value of the constant k.

Answer choices:

A -1

B 1

C 0

D e

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Solution: A

The linear differential equation is already in standard form, so we can identify $P(x)$ and $Q(x)$.

$$P(x) = k$$

$$Q(x) = 0$$

We use $P(x)$ to find the integrating factor.

$$\mu(x) = e^{\int P(x)\ dx}$$

$$\mu(x) = e^{\int k\ dx}$$

$$\mu(x) = e^{kx}$$

Multiply through the differential equation by the integrating factor.

$$e^{kx}\left(\frac{dy}{dx} + ky = 0\right)$$

$$e^{kx}\frac{dy}{dx} + kye^{kx} = 0e^{kx}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(e^{kx}y) = 0$$

then integrate.

$$\int \frac{d}{dx}(e^{kx}y) = \int 0\ dx$$

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$$e^{kx}y = C$$

$$y = Ce^{-kx}$$

Once we have this general solution, we recognize from the initial condition $y(0) = 1$ that $x = 0$ and $y = 1$ and from the initial condition $y(1) = e$ that $x = 1$ and $y = e$, so we'll plug these values into the general solution,

$$1 = Ce^{-k(0)}$$

$$e = Ce^{-k(1)}$$

and then simplify these to find C and k.

$$C = 1$$

$$e = e^{-k}$$

$$k = -1$$

We weren't asked for it, but now we also know that the solution to the differential equation is

$$y = 1e^{-(-1)x}$$

$$y = e^{x}$$

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